LINK TO TINKERCARD :

https://www.tinkercad.com/things/9gCOjN7lWGd/editel?returnTo=%2Fdashboard&sharecode=uvrWaYfNScKWr6u6kPMWembTGBxfLqPrcVqLGWgPbsM

```
const int sensorPin = A0;
const int ledPin = 7;

float voltage;
float current;
float power;
float energy = 0.0;

unsigned long previousTime = 0;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);

  Serial.println("IIoT Energy Monitoring System");
}

void loop() {
  int sensorValue = analogRead(sensorPin);

  // Simulated voltage and current values
  voltage = 230.0;
  current = (sensorValue / 1023.0) * 10.0;

  // Calculate power in watts
  power = voltage * current;

  // Calculate energy in kWh
  unsigned long currentTime = millis();

  if (previousTime != 0) {
    float elapsedHours =
      (currentTime - previousTime) / 3600000.0;

    energy += (power * elapsedHours) / 1000.0;
  }

  previousTime = currentTime;

  Serial.print("Voltage: ");
  Serial.print(voltage);
  Serial.println(" V");

  Serial.print("Current: ");
```

```
  Serial.print(current, 2);
  Serial.println(" A");

  Serial.print("Power: ");
  Serial.print(power, 2);
  Serial.println(" W");

  Serial.print("Energy: ");
  Serial.print(energy, 6);
  Serial.println(" kWh");

  // Optimization alert
  if (power > 1500) {
    digitalWrite(ledPin, HIGH);
    Serial.println("ALERT: High energy consumption!");
    Serial.println("Optimization: Reduce unnecessary loads.");
  } else {
    digitalWrite(ledPin, LOW);
    Serial.println("Energy consumption is normal.");
  }

  Serial.println("-----------------------");
  delay(1000);
}
```

